

7COSC013W: Foundations of AI - Coursework 1 (Project Proposal)

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Forecast-Driven Vehicle Routing for Bike-Share Rebalancing - (A Case Study of the Jersey City & Hoboken Citi Bike Network)

1. Problem Statement and Real-World Context:

Bike-share systems distribute bicycles across a network of docking stations throughout a city for the public's need of transport. A network called Citi Bike, one of the largest such systems in the United States, is a good example of how this can go wrong in practice. Since demand is highly uneven across the day, residential stations tend to empty out during the morning time, while stations near transit and business areas fill up. Due to that, stations frequently end up in one of two failure states: Empty stations (no bikes available to rent), and fully packed stations (no free docks to return a bike).

Normally, companies/ organizations address this through a method called "rebalancing", which means physically redistributing bikes between stations using modes of transport such as vans/trucks. In practice, many bike-share systems rely on rebalancing strategies that are either fixed schedule (the transport follows a predetermined route regardless of that day's actual demand) or reactive (stations are only addressed once they are already empty or full). Both these approaches have clear limitations: a fixed schedule ignores real day-to-day variation in demand, while a reactive approach means the problem has already occurred, and inconvenienced a rider, before anything is done about the issue.

This project addresses that problem by building a predictive rebalancing decision-support system. One that forecasts where and when station imbalance is likely to occur, and uses that forecast to plan an efficient rebalancing route, rather than following a fixed schedule or reacting after the issue has occurred.

2. Proposed AI Technique(s):

This project falls under Category 3: Hybrid Approaches; specifically, the **Search + Learning** sub-category, combining two of the three core technique categories required by the coursework:

- **Machine Learning (Demand Forecasting):** Three regression-based approaches, a Linear Regression model, a Random Forest model, and an Ensemble averaging both, were trained on the historical trip data to predict each station's net bike flow for a given hour. Features used include time of day, day of week, a weekend flag, and station identity. All three methods were compared against a naive baseline (predicting the same value as the same hour, same day of the week, one week earlier), and the best-performing model was selected automatically based on test error rather than assumed in advance, and its predictions were the ones carried forward into the routing stage.

- **Search / Optimization (Rebalancing Route Planning):** Given the forecast, four routing strategies were implemented and compared for a single rebalancing van operating under a fixed time budget: a naive baseline (visiting flagged stations in an arbitrary order), a Priority-Greedy strategy (visiting stations strictly by predicted urgency), a Priority-Weighted Nearest-Neighbor heuristic (balancing urgency against distance at each step), and a Distance-only Nearest-Neighbor heuristic (always travelling to the nearest unvisited flagged station, regardless of urgency). Each was treated as a small-scale vehicle routing problem, evaluated across every relevant hour in the test period rather than a single instance, with the strongest method additionally checked against a brute-force optimal solution on a sample of hours to confirm how close it came to the best possible result.

3. Dataset / Problem Instance Description:

This project used real, publicly available trip-level data from the Citi Bike system data, published under the NYCBS Data Use Policy, which explicitly permits use for academic analysis. Each trip record includes the start and end station/location, station ID, timestamps, and coordinates.

The final scope settled on the Jersey City and Hoboken sub-network (the “JC” data files), narrowed down to the twenty highest-traffic stations by trip volume, over six months of data (January to June 2026). This gave a large enough sample to capture meaningful daily and weekly demand patterns and to properly test the forecasting model on genuinely unseen data, while keeping the problem instance small enough to process, visualize, and interpret within the timeframe of the coursework. Straight-line (Haversine) distance between station coordinates was used for the routing component, rather than real road-network distance, to keep the system self-contained and avoid depending on an external routing API.

4. Success Criteria and Evaluation Metrics:

Success here is defined not as guaranteeing that the proposed method will outperform the naive baseline in every case, but as producing a fair, well-evidenced comparison between naive and proposed approaches, with critical analysis of when and why each performs better or worse.

Specific metrics used were:

- **Forecasting evaluation:** Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) of each model against the naive baseline, measured on a held-out test period the models never saw during training.
- **Routing evaluation:** Total predicted imbalance addressed and total distance travelled by each routing method, aggregated across every flagged hour in the test period rather than judged from a single example, with the best-performing method chosen automatically using a defined selection rule (highest priority addressed, tie-broken by lowest distance).
- **Cross-cutting analysis:** Discussion of trade-offs between the naive and proposed approaches, including the cases where the simpler approach performed competitively, or even better, than a more “sophisticated” one, and what that reveals about the strengths and limits of each technique.